Does typed extension make prose-blocked metrics deterministically computable? A deterministic-tier GRO experiment over 12 compiled ARAs

Abstract

We took 12 already-compiled Agent-Native Research Artifacts (ARAs), generated Grounded Research Object (GRO) deterministic-tier extensions for each — five typed YAML sidecars (`quantities`, `claims_typed`, `refs`, `entities`, `genre`) — and computed the eight deterministic-tier (Tier A) metrics from the GRO spec over the extended shape. The narrow question: *does typing the record turn metrics that were not deterministically computable on the raw prose into pure structural joins, and what do the numbers show across the corpus?* The answer is yes on computability — all eight metrics ran as LLM-free, network-free lookups over the sidecars, and one (`broken_ref_integrity`) caught a real structural defect a prose corpus cannot express. But the result is deliberately narrow: this is the deterministic tier only; the anchored tier (real DOI/registry resolution) and the judged tier (novelty, entailment quality) were not run; the extensions were themselves LLM-generated, which sets a fidelity ceiling; and none of this tests whether the metrics discriminate good science from bad. We report the real per-metric corpus numbers and are explicit about what they do and do not establish.

1. The question

The GRO specification (`IDEAL_FORMAT_SPEC.md`, hereafter "the spec") organizes credibility measurement into three rigor tiers over one id-graph: **Tier A · Deterministic** (pure functions of authored data), **Tier B · Anchored** (external resolution against pinned resolvers), and **Tier C · Judged** (irreducibly semantic verdicts in append-only logs). This experiment concerns Tier A only.

The eight Tier-A metrics share a property that motivated the experiment: on the *original prose ARA* (`PAPER.md` plus its drill-down layers), every one of them is marked `computable_on_prose: false`. The reason in each case is not that the number is hard to compute, but that the field the metric reads *does not exist* until someone types it. A prose claim carries no `claim_type` or `logical_form`; a prose citation is a free-text string, not a record with a `resolvable` boolean; a genre has no `expected_slots` / `present_slots` / `absent_declared` triple. To compute any of these over prose, an LLM would first have to re-extract the exact typed fields the sidecars carry — which means the "metric" would in fact be an LLM annotation step, not a lookup.

So the hypothesis is precise: *if we author those typed fields once as sidecars, the metrics collapse from LLM re-extraction problems into deterministic joins*. This experiment tests that mechanically and reports the corpus numbers that fall out.

2. Method

Corpus. 12 compiled ARAs from the Alzheimer's/dementia research library, spanning clinical biomarker studies (che26, val25, han26, tit26), donanemab trials and secondary analyses (zim25, jes26), single-nucleus/spatial transcriptomics (gau25, ard25), a narrative review (zho25), molecular neuropathology (aki26), a drug-pipeline census (cum26), and a GBD epidemiological analysis (xu25).

Extension phase. For each ARA we wrote five deterministic-tier sidecars into `gro/`: `quantities.yaml` (one canonical typed record per load-bearing number, `Q##`), `claims_typed.yaml` (`C##` with `claim_type`, `logical_form`, `quantity_refs`, `concept_refs`, `proof_refs`, `population_scope`, `population_n`), `refs.yaml` (`R##` with `external_id` and a `resolvable` boolean), `entities.yaml` (`EN` records with `xref`), and `genre.yaml` (`paper_type`, `expected_slots`, `present_slots`, `absent_declared`). Every file was validated with `yaml.safe_load` and passed a cross-reference consistency check (each `quantity_ref` / `concept_ref` / `depends_on` resolves to a real id). The extraction rule throughout was *no guessing*: `not_specified` was used for any field lacking ARA-internal grounding, and honest absences (no printed DOI, no released dataset, no code) were recorded as such rather than filled.

Metrics phase. The eight Tier-A metrics were then computed as pure structural joins over the five sidecars — no LLM calls, no network calls. All 12 ARAs processed; zero failures. During development one real bug surfaced and was fixed: several ARAs (zim25, jes26, xu25, ard25) print negative numbers with the Unicode minus (U+2212), which silently dropped the sign in the reconciliation lexer until normalized — jes26 was misread as 0.667 instead of the correct 1.000. This is itself evidence that the check is doing arithmetic, not string-matching.

3. Results

All eight metrics ran deterministically over all 12 ARAs. Corpus aggregates (from `results.json`):

Metric	Mean	Min	Max	Coverage / notes
quantity_reconciliation_rate	1.000	1.000	1.000	defined on 10/12 ARAs; 2 undefined (null) — no quantity referenced by ≥2 claims
claim_typing_coverage	1.000	1.000	1.000	defined on 12/12
entailment_precondition_rate	0.821	0.400	1.000	defined on 12/12
reference_resolvability_rate	0.374	0.000	1.000	defined on 12/12
entity_anchoring_rate	0.000	0.000	0.000	defined on 12/12; uniformly empty
genre_silent_omissions	0.333 (mean)	−1	4	total = 4; 3/12 ARAs nonzero
overclaim_flags	1.917 (mean)	0	11	total = 23; 3/12 ARAs nonzero
broken_ref_integrity	1.333 (mean)	0	16	total = 16; 1/12 ARAs nonzero

Four findings are worth stating plainly, because they are what the deterministic tier can and cannot see.

The joins compute, and one of them found a real defect. `broken_ref_integrity` caught `ard25`: its `claims.md` cites nine proof references (E01–E09, and E10/E11) with no `logic/experiments.md` — or any experiment ledger — anywhere in the ARA to resolve them against, so all 16 `proof_ref` occurrences across its claims are structurally broken. That ARA's own `genre.yaml` commentary independently documents that several layer files its `PAPER.md` *claims* to contain (experiments, concepts, related_work) do not actually exist on disk. This is a trust signal a prose corpus cannot produce at all: "broken reference" is not a well-formed question until the refs are typed into ids. The same metric returns 0 for the other 11 ARAs, confirming it is not firing spuriously.

`entity_anchoring_rate` is exactly 0.000 across all 12 ARAs and all 176 entities. Every entity carries `xref: not_specified`. This is a *distinct* failure mode from "metric not computable": the L3 entity spine exists as a schema slot and the metric runs, but the compilation pass never populated a single ontology xref. A uniformly-empty typed field is a different — and only now visible — problem than a field that cannot be checked. The metric surfaced it precisely because the field exists to check.

`reference_resolvability_rate` varies from 0.000 to 1.000, exposing pipeline inconsistency. cum26 posts 1.000 (all 10 refs carry printed DOIs); val25, zho25, and ard25 post 0.000 (every ref has `external_id: not_specified`). This spread shows the reference-resolution step of the extension was applied unevenly across the corpus rather than as a uniform stage — some ARAs printed DOIs in their related-work layer, others cited only in-text reference numbers. The 0.374 mean is an honest picture of that unevenness, not a quality score.

`overclaim_flags` concentrates in the two population-scale genres. cum26 (9/10 claims) and xu25 (11/13 claims) account for 20 of the 23 total flags; the third contributor is zho25 (3). These are exactly the pipeline/registry census and GBD epidemiological ARAs, whose claims are typed `population_estimate` with `population_n: not_specified` — the quantifier-inflation pattern the spec's over-claim rule is designed to catch. The other 10 ARAs post zero. This is the metric behaving as specified: it fires on generality-tier claims that omit the sample size the spec promotes to an *expected* slot for that tier (spec axiom 4; L2).

Two ARAs (zho25, aki26) return `null` for `quantity_reconciliation_rate` because neither has a quantity referenced by ≥2 claims, so the "of quantities referenced by ≥2 claims..." denominator is empty. And zho25's `genre_silent_omissions` is **−1** — a genuine internal-consistency artifact: its `genre.yaml` lists `experiments` in `present_slots` but not in `expected_slots`, driving the literal `count(expected) − count(present) − count(absent_declared)` formula negative. We report this rather than clamp it, because it is a real signal that the count-based formula is fragile against a `present_slot` outside the expected set.

4. The honest contrast: extraction vs. join

The load-bearing claim of the experiment is a *before/after* about computability, not a quality verdict. On the raw prose ARA, none of these eight numbers is a lookup. `results.json` records the per-metric reason in each case, and they are all the same shape: the typed field the metric reads simply is not present in prose. `claim_typing_coverage` needs a `claim_type` and `logical_form` per claim — assigning either is LLM inference. `entailment_precondition_rate` needs an explicit `quantity_ref` / `proof_ref` link between a claim and its experiment — that link is inferred, not looked up. `reference_resolvability_rate` needs each citation checked against a resolver — a lookup against the outside world, not a field read. And so on for all eight.

After extension, every one of them is a structural join over `gro/*.yaml`, run with zero LLM and zero network calls, identical run-to-run. That is the whole result on the computability axis: **typing the record moves these metrics from "re-extract with an LLM, then check" to "join and count."** The extension is where the intelligence and the cost live; the metric is then cheap, exact, and reproducible.

This matters because it draws a clean line. Everything the deterministic tier reports is a fact about *the typed record* — its internal consistency, its coverage of its own declared fields, the integrity of its own id-graph. It is not a fact about the paper, and it is not a fact about the world. The next two sections say why.

5. Limitations

These are not caveats bolted on; they are the boundary of what was actually run.

This is the deterministic tier only. The anchored tier (Tier B: L6 reference-spine resolution against a pinned DOI resolver, L7 registry joins against ClinicalTrials.gov/PROSPERO) and the judged tier (Tier C: P1 novelty/significance, P2 typed-rubric entailment) were **not run**. `reference_resolvability_rate` here reflects only whether an `external_id` string was *printed and typed as resolvable* in the sidecar — it does **not** mean any DOI was fetched and confirmed to point at the cited work. Real resolution needs pinned resolvers with index snapshots (spec §Tier B); calibrated judgement needs human calibration sets and judge ensembles (spec §Tier C). Neither exists in this run. Per the spec's own cost accounting, full-density Tier C is $\sim 10^3$–10^4 judge inferences per mid-size GRO and does not run at corpus scale (spec §7, "The Tier-C apparatus does not run at corpus scale").

The extensions were themselves LLM-generated — a fidelity ceiling. Every sidecar was authored by an LLM reading the ARA. This is the spec's central conceded residual (§7, "The compiler selects the content that gets recomputed"): determinism is downstream of a frozen LLM-authored input set, not end-to-end. An extension can *mis-type* — assign the wrong `claim_type`, mark a resolvable ref as `not_specified`, split or merge a quantity, or (as the spec's `vacuity_check` warns) emit a weak/vacuous `logical_form` that deterministically yields "0 contradictions" and high typing coverage while saying nothing. `claim_typing_coverage = 1.000` and `quantity_reconciliation_rate = 1.000` are exact statements about the wrapper, not proof about the paper. In particular, `quantity_reconciliation_rate` is weaker than its name: because each `Q##` is authored once with one canonical value, the check reduces to "is this value rounding-consistent with the numeral in its own authoring quote," not an independent cross-mention reconciliation across abstract/table/discussion prose (that would be a Tier-B/anchored check against `PAPER.md`, not Tier A). The blind-reconstruction passes the spec prescribes to bound this residual (`claim_coverage`, `numeral_reconstruction`, `assumption_coverage`; spec §0.3, §7) were not run here.

No test of whether the metrics discriminate good science from bad. There is no external ground truth in this experiment. We did not label any ARA as strong or weak science and check whether the metrics separate them. This is not an oversight — it is the parent program's central negative result, and it applies in full force here. The v3 metrics tournament concluded, across all 11 winning metric sets and independently across its judges, that "a well-compiled record of bad science and a well-compiled record of good science are, by construction, indistinguishable to every metric in this tournament," because a metric that reads only the artifact "cannot, in principle, verify anything about the world outside that file." Every one of the eight deterministic metrics reported here reads only the `gro/` sidecars. They measure compilation and documentation integrity — internal consistency, self-declared-field coverage, id-graph soundness, resistance to templated padding — and nothing about the correctness, validity, or importance of the underlying findings. A rigorous-looking sidecar over a bad paper scores identically to one over a good paper.

Independence is asserted, not measured, everywhere except one place. Even where the spec's anchored/judged tiers would add a second checker, that checker (outside L3's non-LLM literal matcher, "Arm A") is another LLM whose correlation with the authoring LLM is asserted by construction, not measured (spec §7, "Independence is pipeline-independence, necessary not sufficient"). This experiment ran no second checker at all, so the point is a fortiori: the numbers are self-consistency of a single LLM's output, checked mechanically.

6. Conclusion

Typed extension does what it was hypothesized to do on the narrow axis tested: it converts eight prose-blocked metrics into deterministic structural joins that run corpus-wide with no LLM and no network, identically run-to-run, and it makes at least three things visible that prose cannot express — a structurally broken proof-reference graph (`broken_ref_integrity` on ard25), a uniformly unpopulated entity spine (`entity_anchoring_rate = 0.000`), and a quantifier-inflation pattern concentrated in population-scale genres (`overclaim_flags` on cum26 and xu25). The deterministic tier is cheap, exact, and — importantly — honest about absence: `null`, `not_specified`, and even a negative genre count are reported rather than smoothed away.

But the tier is exactly as narrow as its name. It certifies that a typed record is internally coherent and covers its own declared fields; it does not resolve a single reference against the outside world, does not judge whether any claim is novel or well-supported, and — the finding to keep front of mind — does not and cannot distinguish good science from bad, because it never leaves the artifact. The extensions that make the joins possible are themselves LLM-authored, so even the coherence it certifies is coherence of a wrapper an LLM chose the contents of. What this experiment establishes is a *floor*: the deterministic and anchored tiers can be computed at corpus scale as the cheap gate funders sit on top of. What it does not establish — and what the anchored resolvers, the calibrated judge ensembles, and an external strong/weak ground-truth set would be needed to establish — is whether any of it tracks scientific value at all.

Grounding: metrics from `research/metrics/v4-gro/results.json` and `results.md` (12 ARAs, 0 failures). Tier framing and limitation citations from `research/metrics/v3/tournament/IDEAL_FORMAT_SPEC.md` §1, §7. Parent negative result from `research/metrics/v3/tournament/TOURNAMENT_SUMMARY.md` §1. This paper reports the deterministic tier only; the anchored and judged tiers were not run.